Java Basics

Lab 2

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Announcement

- First assignment will be released in **3/23**! (Due **4/05**)

Announcement

- You should finish the lab practice and submit your job to eTL day before the next lab class starts(3/29 Sun 23:59 PM).
- The answer of the practice will be uploaded after the due.

Overview

- Recap: Java basic review
  - Array
  - if-else
  - for, for-each / while, do-while
- Problem 1 - Student ID Validator
- Problem 2 - TicTacToe

Recap: Array

Main Function

```
String[] cars = {"Volvo", "BMW", "Ford", "Mazda"};
System.out.println(cars[0]);
cars[0] = "Hyundai";
System.out.println(cars[0]);
System.out.println(cars.length);
```

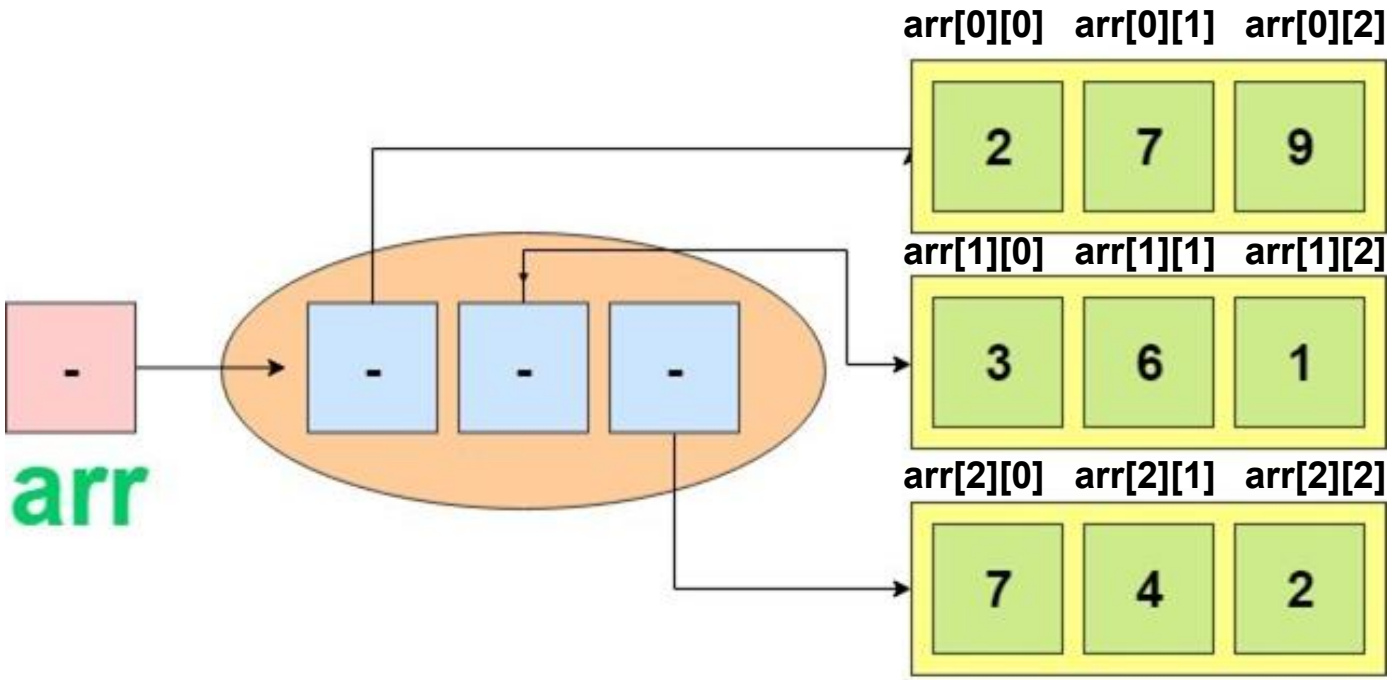
Output

```
Volvo
Hyundai
4
```

Recap: Multi-Dimensional Array

- Each array element is an array

```
int[][] arr = {{2, 7, 9}, {3, 6, 1}, {7, 4, 2}};
```



Recap: Multi-Dimensional Array

Main Function

```
int[][] myNumbers = {{1, 2, 3}, {4, 5}};
for (int row = 0; row < myNumbers.length; ++row) {
    for (int col = 0; col < myNumbers[row].length; ++col) {
        System.out.println(myNumbers[row][col]);
    }
}
```

Output

```
1
2
3
4
5
```

Recap: `if`/`else` Statement

Main Function

```
int time = 22;
if (time < 10) {
  System.out.println("Good morning.");
} else if (time < 20) {
  System.out.println("Good day.");
} else {
  System.out.println("Good evening.");
}
```

Output

```
Good evening.
```

Recap: `for`/`for`-each

Main Function

```
String[] cars = { "Volvo", "BMW", "Ford", "Mazda" };

for (int i = 0; i < 4; i++) {
  System.out.print(cars[i] + " ");
}
System.out.println();

for (String car : cars) { System.out.print(car + " "); }
System.out.println();
```

Output

```
Volvo BMW Ford Mazda
Volvo BMW Ford Mazda
```

Recap: `while`/`do-while`

Main Function

```
int i = 0;
while (i < 5) { System.out.print(i++ + ","); }
System.out.println();

i = 0;
do { System.out.print(i++ + ","); }
while (i < 5);
System.out.println();
```

Output

```
0,1,2,3,4,
0,1,2,3,4,
```

Download the skeleton code!

- Download Lab 2 skeleton.zip file from eTL and unzip it.
- Open the folder as Intellij project. You have to write the codes in the appropriate source files and submit them as lab assignment.

Problem Overview

- Problem 1 - Student ID Checker
  - 1-1 Get String Input (0:05)
  - 1-2 Validation Logic (0:10)
- Problem 2 - TicTacToe
  - 2-1 Get Board Input (0:05)
  - 2-2 Determine Winner (0:10)
  - 2-3 NxN TicTacToe (0:10)

Problem 1 : Student ID Validator

- In StudentIDValidator.java file,
  write a program that checks whether an input string is a valid student ID (XXXX-XXXXX).

5 mins

Problem 1-1 : Get String Input

- Get string input from console and print it.
- Repeat this until the input is “exit”.
- Hint) `Scanner scanner = new Scanner(System.in);`

5 mins

Problem 1-1 : Get String Input

- Example Inputs & Outputs

Console

Input
Output
Input
Output
Input
Output
Input

```
abc
abc
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Computer Programming

2018-12345
2018-12345

exit
```

10 mins

Problem 1-2 : Validation Logic

- Check whether the input string is a valid student ID or not **in the following order**, and print a corresponding message.
  1. The length of the input should be 10.
     -> If not, print `The input length should be 10.`
  2. The 5th character of the input should be '-'.
     -> If not, print `Fifth character should be `-`.`
  3. All characters of the input but the 5th should be digits.
     -> If not, print `Contains an invalid digit.`
  4. If 1~3 are all satisfied, print `XXXX-XXXXX is valid.`

10 mins

Problem 1-2 : Validation Logic

Here's some tips…

- You can use `.charAt` method of String class to get a character at specific index.
- A character can be converted to its Unicode index (https://unicode-table.com/en/). This is useful when checking if a character belongs to specific category.
  - digit: `ch >= '0' && ch <= '9'`
  - non-digit: `ch < '0' || ch > '9'`
  - lowercase alphabet: `ch >= 'a' && ch <= 'z'`
  - uppercase alphabet: `ch >= 'A' && ch <= 'Z'`

10 mins

Problem 1-2 : Validation Logic

- Example Inputs & Outputs (Invalid student IDs)

Console

Input
Output

```
2018-1234
The input length should be 10.
```

Console

Input
Output

```
2018_12345
Fifth character should be ‘-’.
```

Console

Input
Output

```
e018-12345
Contains an invalid digit.
```

10 mins

Problem 1-2 : Validation Logic

- Example Inputs & Outputs (Valid student ID)

Console

Input

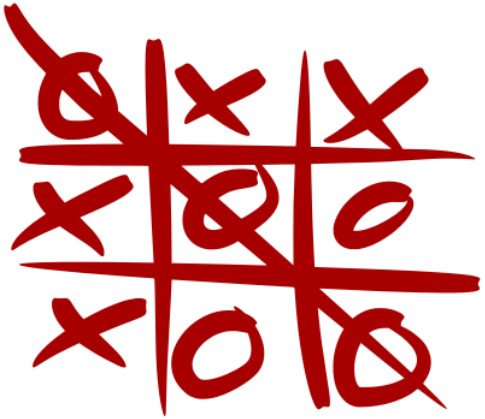
```
2018-12345
```

Output

```
2018-12345 is valid.
```

Problem 2 : TicTacToe

- In TicTacToe.java file, write a program that determines the winner of the Tic-Tac-Toe game given a board.
- Tic-Tac-Toe is a two player game that uses a 3x3 board. Each player alternately places his/her mark on an empty cell of the board.
- The first player who succeeds in placing his/her marks in a horizontal, vertical, or diagonal row is the winner.



5 mins

Problem 2-1 : Get Board Input

- Get 9 integer inputs from the console and save them into a 3x3 integer array.
- Assume each integer of the input is always 0 or 1, which represents the marks of Player 0 or 1.
- You can use the `printBoard` function to check if the board is correctly generated.

Input

```
0 1 0 1 0 0 1 1 1
```

Output when printBoard function called

```
Board:
0 1 0
1 0 0
1 1 1
```

10 mins

Problem 2-2 : Determine Winner

- Determine the winner of the game (who placed his/her marks in a horizontal, vertical, or diagonal row.)
- If Player 0 won, print `Player 0 win.`
- If Player 1 won, print `Player 1 win.`
- If none of them succeeded in making a row, print `Tie.`
- If (1) both of them succeeded in making a row, or (2) the number of marks of two players differ by more than 1, print `Invalid game.`

10 mins

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Problem 2-2 : Determine Winner

0 marks: 8
1 marks: 1

Input
```
0 1 0
1 0 0
1 1 1
```
Output
```
Player 1 win.
```

Input
```
0 0 0
0 0 0
0 0 1
```
Output
```
Invalid game.
```

Input
```
0 1 1
1 0 0
1 1 0
```
Output
```
Player 0 win.
```

Input
```
0 1 1
0 1 1
0 0 1
```
Output
```
Invalid game.
```

Input
```
0 1 0
1 0 0
1 0 1
```
Output
```
Tie.
```

10 mins

Problem 2-3 : NxN TicTacToe

- Extend the game to use NxN board, where N is also an integer input from the console.
- Assume the first input is N, and the rest of the NxN inputs represents the board.
- Assume N >= 3.
- All the rules are same, but now the players have to place N consecutive marks in a horizontal / vertical / diagonal row.

0	1	0	1
1	1	0	1
0	0	1	1
0	0	0	1

10 mins

Problem 2-3 : NxN TicTacToe

Input

```
4
0 1 0 0
1 0 0 1
1 1 1 1
1 0 0 0
```

Output

```
Player 1 win.
```

Input

```
5
0 0 0 0 0
1 1 1 1 1
0 0 0 1 1
1 1 1 0 0
1 0 1 0 0
```

Output

```
Invalid game.
```

Input

```
4
0 1 0 1
1 0 1 0
0 1 0 1
1 0 1 1
```

0 marks: 7
1 marks: 9

Output

```
Invalid game.
```

Submission

- Compress the src folder of the Intellij project into a zip file.
- Rename your zip file as 20XX-XXXXX_{name}.zip - for example, 2026-12345_김세진.zip
- Upload it to eTL - Lab 2 exercise until 3/29 SUN 23:59 PM.
- Your program should contain main function that can be properly executed and print desired outputs.